

STA 630 CURRENT ANSWERS BY TANIYA

Scenario-Based MCQs

1. **Mall Intercept Interview:** A type of survey conducted in shopping centers or malls, where researchers approach people for interviews.
2. **CATI (Computer-Assisted Telephone Interviewing):** A method where interviewers use a computer to assist them in conducting telephone surveys.
3. **Univariate:** Refers to a statistical analysis involving only one variable.
4. **Pretest:** A small-scale trial run of a survey or experiment to identify potential issues.
5. **Snowball Sampling:** A type of non-probability sampling where existing participants refer new participants, creating a "chain" of referrals.
6. **Advantages of Sampling:** Cost, labor, and time savings.
7. **Simple Probability Sampling:** Every member has an equal chance of being selected.
8. **Data Transformation:** Changing data from its original form into a more usable format.
9. **Unidimensional:** All items measure the same construct.
10. **Histogram:** A type of graph used to display the distribution of values for interval or ratio data.
11. **Quasi-Experimental Design:** A research design that resembles an experimental design but lacks random assignment.
12. **Non-reactive Research:** A type of research in which the subjects are unaware of being studied.
13. **Non-reactive Research:** It involves collecting data without the participants' knowledge, ensuring no changes in behavior due to observation.
14. **Counting Vehicles at Traffic:** A type of non-reactive research that collects data without influencing the subjects' behavior

QUESTION/ANSWER

Q1: Difference between Scale and Index

- **Scale:** Measures the intensity or level of something (e.g., strongly agree to strongly disagree). It shows how much of a quality someone has.

- **Index:** Combines multiple questions or factors to create a total score (e.g., an index of economic development based on income, education, and health).

Q3: Type of Question in Scenario

- **Double-barrel question:** It asks two things in one question, making it confusing.
- Example: "Have you ever noticed your employee (something) and reluctant to work for you?"
- This question asks about two things at once, so the respondent may not know which one to answer.

Q4: Why Choose a Personal Interview Instead of a Telephone Interview?

- Poor phone connection.
- Need to see facial expressions and body language.
- Sensitive topic requiring personal interaction.
- Respondent is not comfortable giving detailed answers over the phone.

Q5: Population vs. Sample

- **Population:** All working women in Pakistan.
- **Sample:** 700 women from different organizations in the textile industry.
- The population is the larger group, and the sample is the smaller group taken for study.

Q6: How to Get a Complete Questionnaire? (Any Two Solutions)

1. Follow up with the respondents via call, email, or message to request missing answers.
2. Offer an incentive (small gift, discount, or appreciation) for completing the questionnaire.

Data Representation and Frequency Distribution

- **Data representation:** Refers to organizing data into charts, tables, graphs, etc., to make it easier to understand and interpret.
- **Frequency distribution:** A way to organize and display data to show how often each value or range of values occurs.

Editing and Coding Difference

- **Editing:** Involves reviewing and correcting data for errors or inconsistencies.
- **Coding:** The process of categorizing or labeling responses to prepare data for analysis.

Types of Samples

- **Probability sample:** Every member of the population has a known, non-zero chance of being selected (e.g., simple random sampling, stratified sampling).
- **Non-probability sample:** Members are chosen based on convenience or judgment, and not all members have a chance of being selected (e.g., convenience sampling, snowball sampling).

Disadvantage of Mail Questionnaire

- **Low response rate:** Many people may not fill out and return the questionnaire.
- **Lack of control:** The researcher has no way to clarify questions or ensure correct completion.

Pilot Testing

- **Pilot testing:** Conducting a small-scale trial of the survey or experiment to check the design, data collection process, and overall feasibility before full implementation.

Data Transformation

- **Data transformation:** The process of converting data from its original form into a format that is easier to analyze or interpret.

Limitations of Simple Random Sampling

- **May not be representative:** In some cases, random samples might not represent the population accurately, especially if the sample size is too small.

Why Researchers Prefer Sampling Instead of Collecting Data from Entire Population

- **Cost and time-effective:** Sampling is less expensive and quicker than studying the entire population.

- **Practicality:** Collecting data from everyone in a population can be logistically difficult.

Advantages of Internet Surveys

- **Cost-effective:** No need for printing or physical distribution.
- **Quick response:** Immediate data collection and feedback.
- **Wide reach:** Can be distributed to a large number of people worldwide.

Long Answers

1. Disadvantage of Door-to-Door Interviews

- Time-consuming and expensive.
- Some people may refuse to open the door or answer questions.
- Difficult to cover a large area in a short time.

2. Why Researchers Prefer Smaller Groups Instead of Larger Groups?

- Easier to manage and analyze data.
- Allows deeper discussion and better interaction.
- Saves time and resources.

3. Why Are Field Notes Important in Field Research?

- Help record important observations that might not be in the questionnaire.
- Provide real-life context and deeper understanding.
- Help researchers remember small but important details.

Unit of Analysis in a Statement

To find the unit of analysis, identify the "who" or "what" the research is focusing on.

- Example: If the statement is about studying individuals' behaviors, the unit of analysis is "individuals."
- Example: If the statement is about analyzing the performance of schools, the unit of analysis is "schools."

4. Structured and Unstructured Interview

- **Structured Interview:** Follows a strict set of predetermined questions with fixed response options, making the interview more consistent.
- **Unstructured Interview:** More flexible and open-ended, allowing the interviewer to explore topics in more depth based on the participant's responses.

1. Disadvantages of Self-Administered Interviews

- **Low response rate:** Some people may not complete or return the questionnaire.
- **Misunderstanding of questions:** Without a researcher to clarify, respondents may misinterpret questions.
- **Limited control:** The researcher cannot control the environment or ensure that the participant answers thoughtfully.
- **Increased likelihood of incomplete responses:** Without proper guidance, respondents may leave sections blank.

2. Levels of Editing

- **Editing** refers to the process of checking and correcting data. The levels of editing include:
 - **Primary editing:** Done during data collection, such as checking for errors or incomplete responses.
 - **Secondary editing:** Done after data collection, such as reviewing the data for consistency and completeness.
 - **Final editing:** Ensuring that the data is ready for analysis, with any inconsistencies or errors resolved.

3. Focus Group Discussion (FGD)

- **Focus Group Discussion** is a qualitative research method where a small group of people (usually 6-10) discusses a specific topic or set of issues.
- The researcher moderates the discussion to gather insights, opinions, and ideas.
- **Advantages:**
 - Allows participants to express their views in a group setting, encouraging more ideas.
 - Helps researchers understand the context and reasons behind participants' responses.
 - Can be more engaging than individual interviews.

- **Disadvantages:**

- May be influenced by dominant personalities in the group.
- Can be challenging to analyze due to the richness of the data.
- Not representative of the larger population, as it focuses on a small group.

